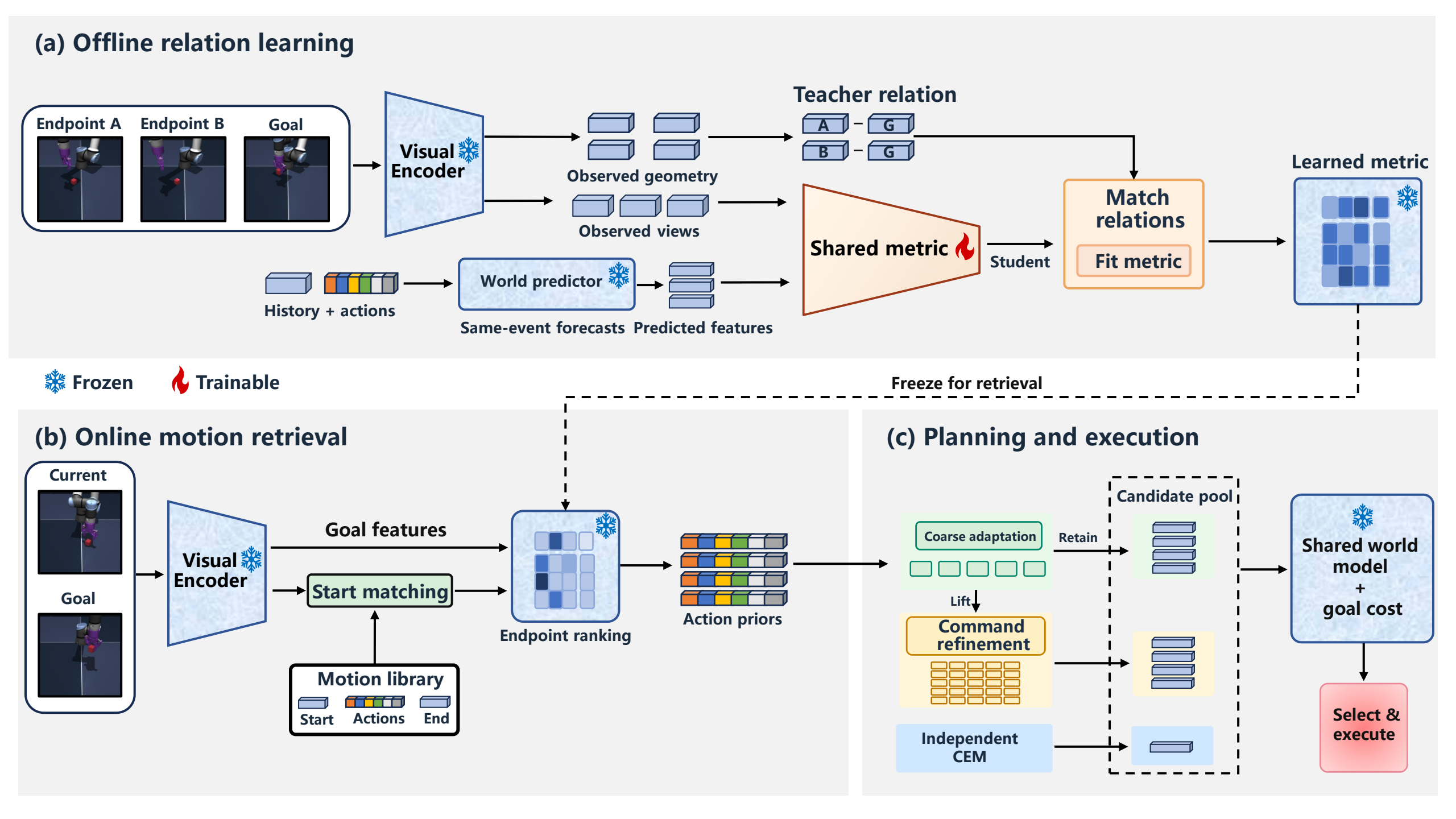

(a) Offline relation learning
Endpoint A
Endpoint B
Goal
Visual Encoder
Observed geometry
Observed views
Teacher relation
A – G
B – G
Shared metric
Student
Match relations
Fit metric
Learned metric
History + actions
World predictor
Same-event forecasts
Predicted features
Frozen
Trainable
Freeze for retrieval
(b) Online motion retrieval
Current
Goal
Visual Encoder
Goal features
Start matching
Motion library
Start
Actions
End
Endpoint ranking
Action priors
(c) Planning and execution
Candidate pool
Coarse adaptation
Retain
Lift
Command refinement
Independent CEM
Shared world model + goal cost
Select & execute